

Experiment 7

Objective: Rollback and Commit commands Create Teacher table with the following fields (Name, DeptNo, Date of joining, DeptName, Location, Salary)

- Insert five records.
- Give Increment of 25% salary for Mathematics Department.
- Perform Rollback command.
- Give increment of 15% salary for Commerce Department.
- Perform Commit command.

```
CREATE DATABASE IF NOT EXISTS school_management;  
USE school_management;  
SET SQL_SAFE_UPDATES = 0;
```

```
DROP TABLE IF EXISTS Teacher;  
CREATE TABLE Teacher (  
    TeacherName VARCHAR(50),  
    DeptNo INT,  
    DateOfJoining DATE,  
    DeptName VARCHAR(50),  
    LocationName VARCHAR(50),  
    SalaryAmt INT  
);
```

(a) Insert five records.

```
INSERT INTO Teacher (TeacherName, DeptNo, DateOfJoining, DeptName, LocationName, SalaryAmt)  
VALUES  
( 'amit kumar', 10, '2021-01-10', 'mathematics', 'delhi', 40000),  
( 'sunita devi', 20, '2020-03-15', 'commerce', 'mumbai', 35000),  
( 'rahul sharma', 10, '2022-05-22', 'mathematics', 'delhi', 42000),  
( 'priya nair', 30, '2019-11-01', 'science', 'kochi', 50000),  
( 'aniket das', 20, '2023-02-14', 'commerce', 'kolkata', 32000);
```

(b) Give Increment of 25% salary for Mathematics Department.

```
START TRANSACTION;  
UPDATE Teacher
```

	TeacherName	DeptNo	DateOfJoining	DeptName	LocationName	SalaryAmt
▶	amit kumar	10	2021-01-10	mathematics	delhi	50000
	rahul sharma	10	2022-05-22	mathematics	delhi	52500

```
SET SalaryAmt = SalaryAmt * 1.25
WHERE DeptName = 'mathematics';
SELECT * FROM Teacher WHERE DeptName = 'mathematics';
```

(c) Perform Rollback command.

```
ROLLBACK;
```

```
SELECT * FROM Teacher WHERE DeptName = 'mathematics';
```

	TeacherName	DeptNo	DateOfJoining	DeptName	LocationName	SalaryAmt
▶	amit kumar	10	2021-01-10	mathematics	delhi	40000
	rahul sharma	10	2022-05-22	mathematics	delhi	42000

(d) Give increment of 15% salary for Commerce Department.

```
START TRANSACTION;
UPDATE Teacher
SET SalaryAmt = SalaryAmt * 1.15
WHERE DeptName = 'commerce';
```

(e) Perform Commit command.

```
COMMIT;
```

```
SELECT *
FROM Teacher;
```

	TeacherName	DeptNo	DateOfJoining	DeptName	LocationName	SalaryAmt
▶	amit kumar	10	2021-01-10	mathematics	delhi	40000
	sunita devi	20	2020-03-15	commerce	mumbai	40250
	rahul sharma	10	2022-05-22	mathematics	delhi	42000
	priya nair	30	2019-11-01	science	kochi	50000
	aniket das	20	2023-02-14	commerce	kolkata	36800

```
SETSQL_SAFE_UPDATES=1;
```

Experiment 8

Create Sales table with the following fields (Sales No, Salesname, Branch, Salesamount, DOB) a) Insert five records.

- b) Calculate total salesamount in each branch.
 - c) Calculate average salesamount in each branch.
 - d) Display all the salesmen, DOB who are born in the month of December as day in character format i.e. 21-Dec-09.
-
- e) Display the name and DOB of salesman in alphabetical order of the month.

```
CREATE DATABASE IF NOT EXISTS sales_db; USE
sales_db;
```

```
DROP TABLE IF EXISTS Sales;
CREATE TABLE Sales (
    SalesNo INT PRIMARY KEY,
    SalesName VARCHAR(50),
    BranchName VARCHAR(50),
    SalesAmount INT,
    DateOfBirth DATE
);
```

(a) Insert five records.

```
INSERT INTO Sales (SalesNo, SalesName, BranchName, SalesAmount, DateOfBirth) VALUES
(101, 'arnav gupta', 'north', 45000, '1995-12-21'),
(102, 'sana sheikh', 'south', 32000, '1998-05-15'),
(103, 'kabir malhotra', 'north', 55000, '1996-12-09'),
(104, 'isha patel', 'east', 40000, '1997-12-05'),
(105, 'vihaan nair', 'south', 28000, '1999-08-22');
```

(b) Calculate total salesamount in each branch.

```
SELECT BranchName, SUM(SalesAmount) AS TotalSales FROM
Sales
```

GROUPBYBranchName;

	BranchName	TotalSales
▶	north	100000
	south	60000
	east	40000

(c) Calculate average salesamount in each branch.

AIM:

```
SELECT BranchName, AVG(SalesAmount) AS AverageSales
FROM Sales
GROUP BY BranchName;
```

	BranchName	AverageSales
▶	north	50000.0000
	south	30000.0000
	east	40000.0000

(d)Display all the salesmen, DOB who are born in the month of December as day in characterformat i.e. 21-Dec-09.

```
SELECT SalesName, DATE_FORMAT(DateOfBirth, '%d-%b-%y')
AS FormattedDOB FROM
Sales
WHERE MONTH(DateOfBirth)=12;
```

	SalesName	FormattedDOB
▶	arnav gupta	21-Dec-95
	kabir malhotra	09-Dec-96
	isha patel	05-Dec-97

(e) Display the name and DOB of salesman in alphabetical order of the month.

```
SELECT SalesName, DateOfBirth
FROM Sales
ORDER BY MONTHNAME(DateOfBirth) ASC;
```

	SalesName	DateOfBirth
▶	vihaan nair	1999-08-22
	arnav gupta	1995-12-21
	kabir malhotra	1996-12-09
	isha patel	1997-12-05
	sana sheikh	1998-05-15

AIM:

Experiment 9

Consider the following tables namely “DEPARTMENTS” and “EMPLOYEES” Their schemas are as follows:

Departments (dept no , dept name , dept_location);

Employees (emp_id , emp_name , emp_salary,dept_no);

- a) Develop a query to grant all privileges of employees table into departments table.
- b) Develop a query to grant some privileges of employees table into departments table.
- c) Develop a query to revoke all privileges of employees table from departments table.
- d) Develop query to revoke some privileges of employees table from departments table.
- e) Write a query to implement the save point.

```
CREATE DATABASE IF NOT EXISTS company_records; USE
company_records;
```

```
CREATE TABLE IF NOT EXISTS Departments (
    DeptNo INT PRIMARY KEY,
    DeptName VARCHAR(50),
    DeptLocation VARCHAR(50)
);
```

```
CREATE TABLE IF NOT EXISTS Employees (
    EmpId INT PRIMARY KEY,
    EmpName VARCHAR(50),
    EmpSalary INT,
    DeptNo INT,
    FOREIGN KEY (DeptNo) REFERENCES Departments(DeptNo)
);
```

```
CREATE USER IF NOT EXISTS 'dept_manager'@'localhost' IDENTIFIED BY 'secure_pass123';
```

(a)Develop a query to grant all privileges of employees table into departments table.

```
GRANT ALL PRIVILEGES ON company_records.Employees TO 'dept_manager'@'localhost';
```

AIM:

(b)Develop a query to grant some privileges of employees table into departments table.

```
GRANT SELECT, UPDATE ON company_records.Employees TO 'dept_manager'@'localhost';
```

(c) Develop a query to revoke all privileges of employees table from departments table.

```
REVOKE ALL PRIVILEGES ON company_records.Employees FROM 'dept_manager'@'localhost';
```

(d)Develop a query to revoke some privileges of employees table from departments table.

```
REVOKE UPDATE ON company_records.Employees FROM 'dept_manager'@'localhost'; (e)
```

Write a query to implement the save point.

```
START TRANSACTION;  
  
INSERT INTO Employees (EmpId, EmpName, EmpSalary, DeptNo)  
VALUES (501, 'rajat verma', 45000, NULL);  
SAVEPOINT entry_alpha;  
UPDATE Employees SET EmpSalary = 48000 WHERE EmpId = 501; COMMIT;
```

	user	host
▶	dept_manager	localhost
	mysql.infoschema	localhost
	mysql.session	localhost
	mysql.sys	localhost
	root	localhost
	user1	localhost

	Grants for dept_manager@localhost
▶	GRANT USAGE ON *.* TO `dept_manager` @`l... GRANT SELECT ON `company_records`.`emplo...

AIM:

Experiment 10

Using the tables “DEPARTMENTS” and “EMPLOYEES” from Lab 9 perform the following queries:

- a) Display the employee details, departments that the departments are same in both the emp and dept.
- b) Display the employee name and Department name by implementing a left outer join.
- c) Display the employee name and Department name by implementing a right outer join.
- d) Display the details of those who draw the salary greater than the average salary.

```
CREATE DATABASE IF NOT EXISTS company_joins; USE
company_joins;
```

```
CREATE TABLE IF NOT EXISTS Departments (
    DeptNo INT PRIMARY KEY,
    DeptName VARCHAR(50),
    DeptLocation VARCHAR(50)
);
```

```
CREATE TABLE IF NOT EXISTS Employees (
    EmpId INT PRIMARY KEY,
    EmpName VARCHAR(50),
    EmpSalary INT, DeptNo
    INT
);
```

```
INSERT INTO Departments (DeptNo, DeptName, DeptLocation) VALUES
(10, 'mathematics', 'block a'),
(20, 'commerce', 'block b'),
(30, 'science', 'block c'),
(40, 'humanities', 'block d');
```

```
INSERT INTO Employees (EmpId, EmpName, EmpSalary, DeptNo) VALUES
(701, 'amit kumar', 45000, 10),
(702, 'sunita devi', 55000, 20),
(703, 'rahul sharma', 38000, 10),
(704, 'priya nair', 60000, 30),
(705, 'vicky singh', 42000, NULL);
```

(a) Display the employee details, departments that the departments are same in both the emp and dept.

AIM:

```
SELECT E.*, D.DeptName, D.DeptLocation  
FROM Employees E
```

```
JOIN Departments D ON E.DeptNo=D.DeptNo;
```

	EmpId	EmpName	EmpSalary	DeptNo	DeptName	DeptLocation
▶	701	amit kumar	45000	10	mathematics	block a
	702	sunita devi	55000	20	commerce	block b
	703	rahul sharma	38000	10	mathematics	block a
	704	priya nair	60000	30	science	block c

(b) Display the employee name and Department name by implementing a left outer join.

```
SELECT E.EmpName, D.DeptName  
FROM Employees E  
LEFT OUTER JOIN Departments D ON E.DeptNo = D.DeptNo;
```

	EmpName	DeptName
▶	amit kumar	mathematics
	sunita devi	commerce
	rahul sharma	mathematics
	priya nair	science
	vicky singh	NULL

(c) Display the employee name and Department name by implementing a right outer join.

```
SELECT E.EmpName, D.DeptName  
FROM Employees E  
RIGHT OUTER JOIN Departments D ON E.DeptNo = D.DeptNo;
```

	EmpName	DeptName
▶	rahul sharma	mathematics
	amit kumar	mathematics
	sunita devi	commerce
	priya nair	science
	NULL	humanities

(d) Display the details of those who draw the salary greater than the average salary.

```
SELECT * FROM Employees  
WHERE EmpSalary > (SELECT AVG(EmpSalary) FROM Employees);
```

	EmpId	EmpName	EmpSalary	DeptNo
▶	702	sunita devi	55000	20
	704	priya nair	60000	30
•	NULL	NULL	NULL	NULL